

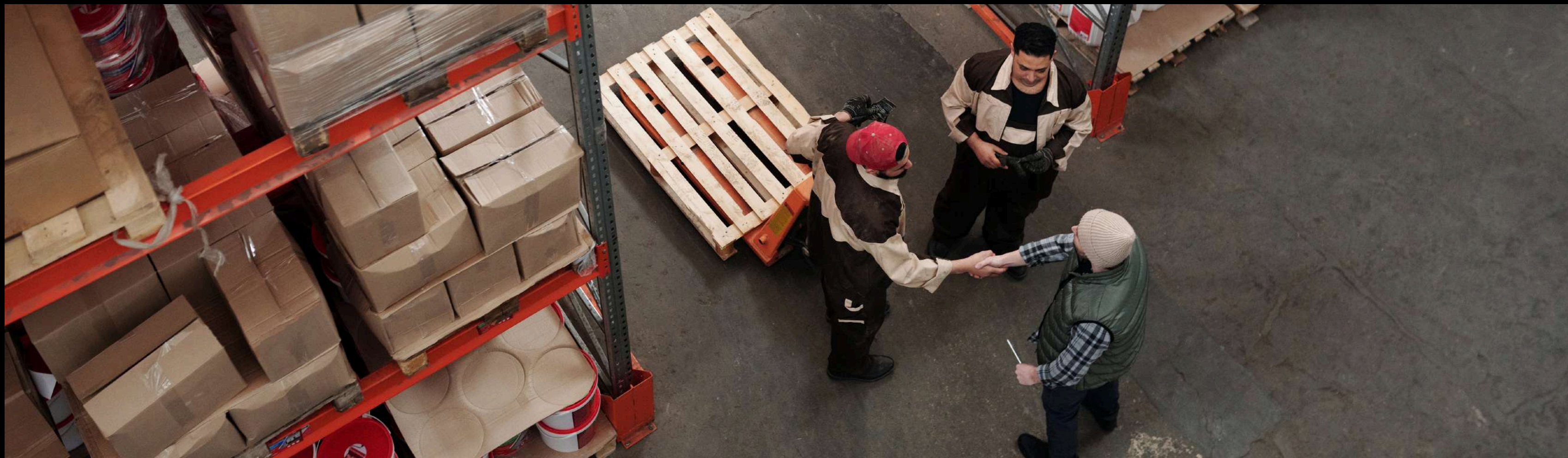
IBM BOB HACKATHON



PORTPULSE

CONTAINER CONGESTION PREDICTOR &
PORT OPERATIONS OPTIMISER

An Overview of the
solution prepared by
PC Pirates



PORTS ARE FLYING BLIND UNTIL IT'S TOO LATE

THE PROBLEM

The Challenge

Build a Bob solution that predicts congestion hotspots using vessel schedules and berth capacity data, recommends alternate routing strategies, optimises berth and crane assignments, and generates a 72-hour port operations plan for shift supervisors.

- 2021 LA/Long Beach backlog: 100+ ships waiting offshore for weeks
- Global economic impact: \$10B+ in supply chain losses
- Root cause: berths, cranes, and yard space are still allocated manually in spreadsheets
- Congestion is spotted reactively — after vessels are already queuing
- Rerouting decisions arrive too late to matter

Why Port Delays Matter?

The cost of congestion starts long before a vessel reaches the berth.



A SINGLE VESSEL CAN INCUR

\$25,000 – \$75,000 / day

IN DEMURRAGE PENALTIES

Every idle hour at anchorage waiting for a berth burns thousands of dollars in direct contractual costs.



ETA
Uncertainty



Berth
Conflict



Anchorage
Idle Time



Cost +
Congestion



APPLYING SLOW-STEAMING
CAN SAVE UP TO

14%

FUEL CONSUMPTION
PER VOYAGE

Reducing hurry-up-and-wait not only cuts costs (~\$600/ton) but also lowers CO₂ emissions.

**IDLE TODAY.
HIGHER COSTS
TOMORROW.**

01 | COST IMPACT



\$25K – \$75K

Demurrage / day

Idle vessels burn thousands in contractual penalties.

Source: BIMCO
(Baltic & International Maritime Council)

02 | FUEL & EMISSIONS



Up to 14%

Fuel waste per voyage

Hurry-up-and-wait leads to avoidable fuel and CO₂ waste (~\$600/ton).

Source: IMO
(International Maritime Organization – JIT Guide)

03 | SCHEDULE RELIABILITY



1 in 3

Container vessels
off-schedule

Carrier-reported ETAs alone cannot guarantee smooth berth planning.

Source: Sea-Intelligence
(Global Liner Performance)

04 | FLEET CAPACITY



8% – 10%

Global fleet capacity
immobilized

Congested anchorages trap billions of dollars in productive fleet assets.

Source: UNCTAD
(Review of Maritime Transport)

05 | PORT EFFICIENCY



>30%

Port call time is
unproductive

Most delays occur before the vessel even reaches the berth.

Source: World Bank & S&P
(Global Container Port Performance Index)



The Bigger Picture

Uncertainty at sea creates real costs on land — higher fuel use, idle assets, congested ports, and unreliable schedules.
Solving port delays isn't just operational — it's economic, environmental, and global.



**EFFICIENT PORTS
STRONGER TOMORROWS**

Pain Point

What Happens Today

Reactive Planning

Congestion is noticed only after vessels start queuing.

Uncertain ETAs

Delays and schedule changes quickly make berth plans outdated.

Manual Berth & Crane Allocation

Human errors can cause overlaps, unsafe assignments, and resource waste.

Cascading Bottlenecks

One berth or crane delay can disrupt the following operations.



OUR SOLUTION

→ INTRODUCING PORTPULSE

- A PREDICTIVE, ROLE-BASED PORT OPERATIONS PLATFORM THAT TURNS REACTIVE FIREFIGHTING INTO PROACTIVE PLANNING
- LIVE AT: PORTPULSE
- COMBINES FORECASTING, PRESCRIPTIVE RECOMMENDATIONS, AND REAL-TIME OPERATIONAL TOOLS IN ONE DASHBOARD
- BUILT FOR THE PEOPLE WHO ACTUALLY RUN THE PORT FLOOR — NOT JUST ANALYSTS



THE ASK

THE CHALLENGE DEMANDS

- PREDICT CONGESTION HOTSPOTS FROM VESSEL SCHEDULES + BERTH CAPACITY DATA
- RECOMMEND ALTERNATE ROUTING STRATEGIES
- OPTIMISE BERTH AND CRANE ASSIGNMENTS
- GENERATE A 72-HOUR PORT OPERATIONS PLAN FOR SHIFT SUPERVISORS

PortPulse v3

SMARTER PORTS. SMOOTHER TOMORROWS.

PREDICT | OPTIMIZE | OPERATE

*Efficient
Ports
Brighter
Tomorrow*



STAND-OUT FEATURES

PortPulse v3

Smarter Decisions. Safer Ports. Brighter Tomorrow.

1



Guaranteed Operational Safety

Hard constraints enforce draft, length and non-overlap. No impossible dockings.

2



Proven Predictive Accuracy

ML ETA correction outperforms naive baseline.

3



Proactive, Not Reactive

72-hour, 720-slot horizon with early warnings and slow-steaming advisories.

4



Trust Through Explainability

Driver breakdowns and calibrated confidence intervals show why a berth is risky.

5



Human in Control

All recommendations wait for approval. Accept, Modify or Reject — fully audited.

6



Always Fresh

Event-driven re-prediction. Cached responses in ~11 ms (from ~5.8 s).

7



Fast at Scale

Auto-optimizer in ~6 s. Batch commit of 50 assignments in ~4.5 s (from 37 s).

8



Four AI Disciplines in One Product

Predictive ML, prescriptive optimisation, generative AI and governance.

9



Plug-in, Not Rip-and-Replace

Works alongside existing TOS instead of a multi-million dollar, 12-24 month replacement.

10



Secure by Design

JWT, bcrypt, server-side RBAC, audit trail, sanitised AI.

WHY PORTPULSE IS >>>

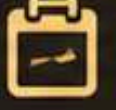
Competitive Landscape

Different strengths. A more intelligent future.

*Smarter Decisions
Busier Ports
Better Tomorrow*

PortPulse v3

PREDICT | OPTIMIZE | OPERATE

 Feature	 Legacy TOS (e.g., Navis N4)	 project44	 Portchain	 PortPulse (Our Solution) 
 Operational Stance	Reactive Focuses on terminal execution and current operations	Informational Visibility into vessel and supply-chain status	Planning-focused Berth planning and schedule alignment	Predictive & Prescriptive Predicts 72h bottlenecks and recommends corrective actions
 ETA Handling	Uses vessel/carrier ETA within terminal planning workflows	Real-time vessel tracking and ETA visibility	Uses vessel schedules and ETA information for berth planning	ML-based ETA correction + continuous updates for more reliable planning
 Scheduling Engine	TOS-based operational planning and rules	Not primarily a berth/crane scheduling optimizer	Digital berth planning + resource allocation	MILP optimization for berth, crane, yard and vessel sequencing
 Operator Assistance	Operational interfaces and planning tools	Dashboards & alerts	Planning and scenario-analysis tools	GenAI Copilot: Natural-language queries + automated shift briefings
 Cost & Deployment	Enterprise-grade, complex integration and deployment	SaaS/API-based visibility solution	Specialized berth-planning platform	Agile & modular: Designed to plug into existing TOS/data infrastructure without replacing heavy hardware

 **Predict Risks Early** |  **Optimize Resources** |  **Assist Operators** |  **Enable Sustainable Growth**

PORTPULSE
A SMARTER PORT
IS A STRONGER TOMORROW



OUR SCORECARD



1. Operational Delay Detection (Classification $\geq 1.0h$)

Identify vessels with significant operational delays (≥ 1 hour)



Goal:
Catch real delays, avoid false alarms,
and keep on-time vessels clear.

Metric	Score	Naive Baseline	Improvement / Interpretation
Accuracy	95.88%	34.4%	+61.48% improvement over baseline
Precision	99.18%	34.4%	~0.8% false positive rate (near-zero false alarms)
Recall (Sensitivity)	89.30%	100.0%	Identifies 89.3% of all genuine delay disruptions
Specificity	98.98%	0.0%	Accurately clears on-time vessels
F1-Score	0.940	0.512	Harmonic balance between Precision and Recall
ROC-AUC Score	0.946	0.500	Outstanding threshold discrimination



2. Continuous Arrival & Dwell Prediction (Regression)

Predict vessel arrival times and port dwell durations with high accuracy



Goal:
More accurate predictions $\rightarrow$ better planning,
less congestion, higher efficiency.

Metric	Naive Baseline	PortPulse Model v3	Performance Gain / Interpretation
Mean Absolute Error (MAE) Average prediction error	1.78 hours	0.66 hours (~39.6 min)	+63.2% Gain
Root Mean Squared Error (RMSE) Penalizes larger errors	2.18 hours	1.16 hours	+46.6% Error Reduction
Goodness of Fit (R^2) Explains variance in real dwell time	0.000	0.713	Explains >71% of real dwell variance
Residual Uncertainty (σ) Uncertainty in predictions	± 2.18 hours	± 0.45 hours	Calibrated 80% confidence interval

SECURITY

AS PIRATES CAN BE DIGITAL TOO!

AUTHENTICATION & ACCESS CONTROL

- MFA FOR ADMIN & TERMINAL MANAGER LOGINS
- RBAC ENFORCED AT API LEVEL, NOT JUST UI
- SESSION TIMEOUT / AUTO-LOGOUT ON INACTIVITY

DATA PROTECTION

- ENCRYPTION AT REST (MASTER DATA) & IN TRANSIT (HTTPS/TLS)
- FIELD-LEVEL MASKING IN CSV EXPORTS BASED ON ROLE

AUDIT & INTEGRITY

- TAMPER-EVIDENT (HASH-CHAINED) ACTIVITY LOG
- ACTION-LEVEL LOGGING: USER ID, TIMESTAMP, IP ON EVERY WRITE

INPUT & API SECURITY

- INPUT VALIDATION/SANITIZATION ON ALL FORMS
- RATE LIMITING ON KEY ENDPOINTS (ASK AI, AUTO SYNC)
- CSRF PROTECTION ON STATE-CHANGING REQUESTS

ADMIN CONTROLS

- SECOND-ADMIN APPROVAL FOR CRITICAL ACTIONS
- LEAST-PRIVILEGE DEFAULTS FOR NEW USERS

MONITORING

- ANOMALY DETECTION ON ACTIVITY LOG (UNUSUAL PATTERNS, NEW-LOCATION LOGINS)
- REAL-TIME SECURITY ALERTING ON DASHBOARD

OUR SOLUTION IS NOT JUST ACCURATE,
IT IS SECURE!



ROLE-BASED ACCESS

ONE PLATFORM, EVERY STAKEHOLDER

- *ADMINISTRATOR — MANAGES USERS, MASTER DATA, AND SYSTEM CONFIGURATION*
- *SHIFT SUPERVISOR — EXECUTES THE 72-HOUR OPERATIONS PLAN*
- *VESSEL PLANNER — MANAGES SCHEDULES AND ROUTING DECISIONS*
- *TERMINAL MANAGER — OVERSEES BERTH, CRANE, AND YARD OPERATIONS*
- *EACH ROLE SEES A TAILORED VIEW — RIGHT INFORMATION, RIGHT PERSON, RIGHT TIME*



CONGESTION PREDICTION & 72-HOUR OPERATIONS PLAN

SEEING THE BACKLOG BEFORE IT FORMS

- CONGESTION HEATMAP — VISUALISES HOTSPOT RISK ACROSS BERTHS AND YARD ZONES IN REAL TIME
- PREDICTIVE MODEL FLAGS HIGH-RISK WINDOWS USING VESSEL SCHEDULES AND BERTH CAPACITY DATA
- SHIFTS DECISION-MAKING FROM REACTIVE TO ANTICIPATORY

FROM PREDICTION TO ACTION

- AUTO-GENERATED 72-HOUR PLAN TAILORED FOR SHIFT SUPERVISORS
- TRANSLATES CONGESTION FORECASTS INTO A CONCRETE OPERATIONAL SCHEDULE
- REMOVES THE MANUAL SPREADSHEET PLANNING BOTTLENECK AT THE CORE OF THE PROBLEM STATEMENT





MONITORING & ANALYTICS

KEEPING OPERATIONS ACCOUNTABLE

- *ACTIVITY LOG — FULL AUDIT TRAIL OF ACTIONS AND CHANGES ACROSS THE PLATFORM*
- *BENCHMARKS — PERFORMANCE TRACKING AGAINST OPERATIONAL TARGETS/HISTORICAL BASELINES*
- *AUTO SYNC — KEEPS VESSEL, BERTH, AND SCHEDULE DATA CONTINUOUSLY UP TO DATE*

USABILITY & DATA MANAGEMENT

- *GUIDED TOUR — IN-APP WALKTHROUGH THAT DOUBLES AS A LIVE USER MANUAL*
- *DARK MODE / LIGHT MODE — ACCESSIBLE FOR ROUND-THE-CLOCK PORT OPERATIONS*
- *MASTER DATA MANAGEMENT — CENTRALISED CONTROL OF CORE REFERENCE DATA*
- *ADMIN CAN CREATE AND MANAGE NEW USERS*
- *EXPORT TO CSV AND PRINT SUPPORT — FOR REPORTING, COMPLIANCE, AND OFFLINE USE*

SIMULATION & DECISION SUPPORT

TESTING DECISIONS BEFORE COMMITTING

- **DELAY SIMULATOR** — MODELS THE DOWNSTREAM IMPACT OF VESSEL OR BERTH DELAYS BEFORE THEY HAPPEN
- **LETS PLANNERS STRESS-TEST "WHAT-IF" SCENARIOS AHEAD OF REAL DISRUPTION**
- **ASK AI** — CONVERSATIONAL ASSISTANT FOR QUICK, CONTEXTUAL OPERATIONAL QUERIES

[📍 Quayside Spatial Map](#) [☰ Berthing Manifest Table](#) [📅 72h Gantt Timeline](#) [⚡ Congestion Testing Lab](#) [🖨️ AI Shift Briefing](#)



Operational Congestion Testing & Shock Lab

Inject real-world disruption events into the port pipeline to test how our machine learning models forecast congestion and trigger mitigation actions

[🧠💡 How Predictions Work & Where to See Them](#)



Mega-Ship Surge

Clusters 3 Ultra-Large Container Vessels (ULCVs) into an identical 3-hour arrival window to overload quays and test anchorage stacking.

[⚡ Inject Mega-Ship Surge](#)



STS Crane Outage

Takes down STS Gantry Crane #2 on Berth 02, slashing discharge throughput by 50% and doubling ship dwell time.

[⚠️ Inject Crane Breakdown](#)



Low Tide Anomaly

Drops fairway channel draft limits by 2.5m, restricting vessels with draft > 13.0m from entering quayside until next tide cycle.

[💧 Inject Low Tide Anomaly](#)



Restore Baseline

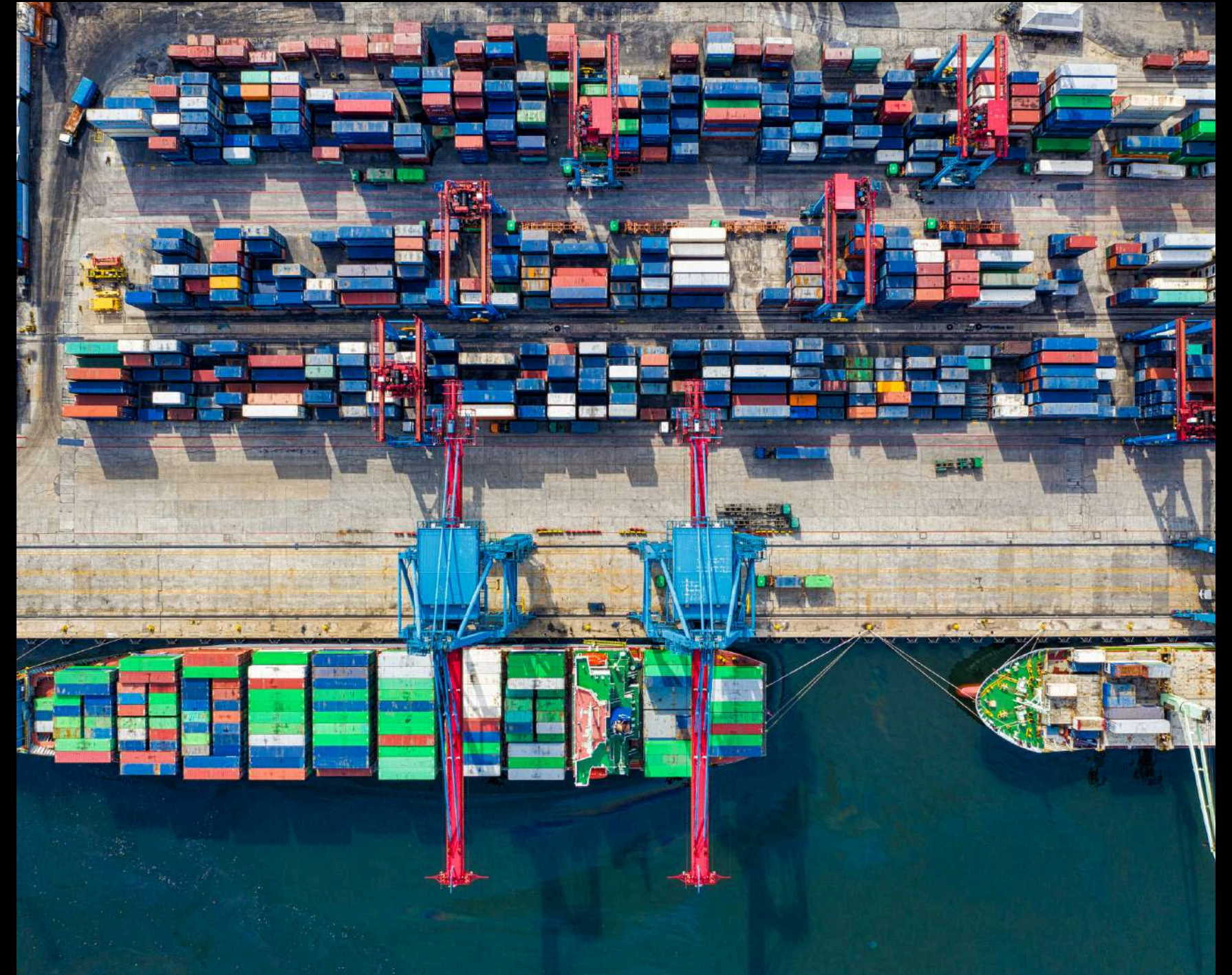
Flushes all active shocks and restores the calibrated 50-vessel baseline dataset and optimal solver schedules.

[🔄 Reset to Baseline](#)

BERTH & RESOURCE OPTIMISATION

OPTIMISING THE PHYSICAL OPERATION

- *BERTH ALLOCATOR — INTELLIGENTLY ASSIGNS BERTHS ACROSS VESSELS TO MINIMISE CONGESTION*
- *TERMINAL MAP — VISUAL, SPATIAL VIEW OF TERMINAL LAYOUT AND VESSEL/YARD STATUS*
- *LIVE QUEUE — REAL-TIME TRACKING OF VESSELS WAITING OR IN PROCESS*
- *TOGETHER, THESE REPLACE MANUAL SPREADSHEET-BASED BERTH/CRANE ALLOCATION*



PRESCRIPTIVE ACTIONS

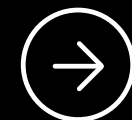
NOT JUST ALERTS — RECOMMENDATIONS

- *SYSTEM DOESN'T STOP AT FLAGGING RISK — IT PRESCRIBES SPECIFIC CORRECTIVE ACTIONS*
- *SUGGESTS ALTERNATE ROUTING STRATEGIES TO AVOID PREDICTED BOTTLENECKS*
- *CONVERTS INSIGHT INTO AN EXECUTABLE NEXT STEP FOR OPERATORS*



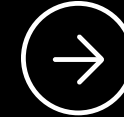
IMPACT

- CONVERTS WEEKS OF REACTIVE CONGESTION RESPONSE INTO HOURS OF PROACTIVE PLANNING
- DIRECTLY TARGETS THE MANUAL, SPREADSHEET-DRIVEN ALLOCATION PROCESS THAT CAUSED THE 2021 BACKLOG
- SCALABLE ACROSS PORTS OF ANY SIZE — SMALL TERMINALS TO MAJOR HUBS LIKE LA/LONG BEACH
- REDUCES VESSEL WAIT TIME, DEMURRAGE COSTS, AND DOWNSTREAM SUPPLY CHAIN DISRUPTION



FRONTEND

- CORE LIBRARY: REACT 18
- LANGUAGE: TYPESCRIPT
- BUILD TOOL: VITE
- STYLING: TAILWIND CSS
- MAPPING UI: LEAFLET AND REACT-LEAFLET (FOR INTERACTIVE MAPS)
- ICONS: LUCIDE REACT



BACKEND

- FRAMEWORK: FASTAPI (PYTHON) RUNNING ON UVICORN
- DATABASE ORM: SQLALCHEMY
- DATABASE: POSTGRESQL (INDICATED BY PSYCOPG2-BINARY) ALONG WITH SQLITE FOR LOCAL DEVELOPMENT (PORTPULSE.DB)
- DATA VALIDATION & SETTINGS: PYDANTIC
- AUTHENTICATION: JWT & PASSWORD HASHING USING PYTHON-JOSE AND PASSLIB
- DATA SCIENCE / MACHINE LEARNING: NUMPY, SCIPY, AND SCIKIT-LEARN (LIKELY USED FOR THE CONGESTION PREDICTION AND OPTIMIZATION LOGIC)
- TESTING: PYTEST AND PYTEST-ASYNCIO

TECH HIGHLIGHTS



THANK YOU



LIVE DEMO: [CLICK HERE](#)

THE PEOPLE BEHIND IT

- Naitik Patel
- Shah Diya Rajan
- Brij Patel
- Dodiya Rudra Ketansinh

